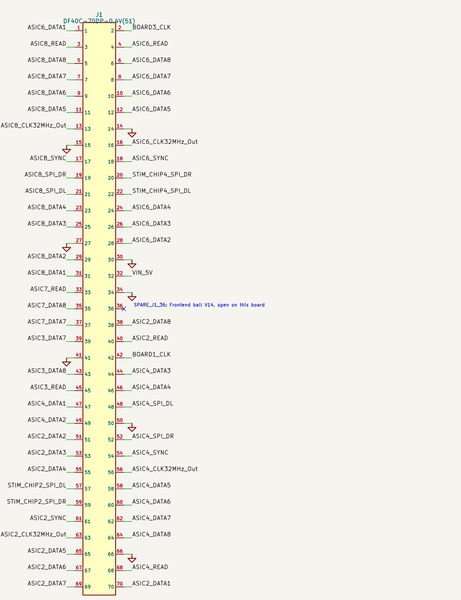
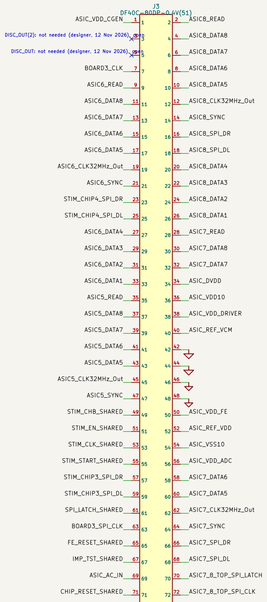
FPGA–LDO PCB SCHEMATIC V35 – OPTION A, ONE ROUTER/LDO BOARD (PIN MAP V6, +/–8 V STIMULATION RAILS)

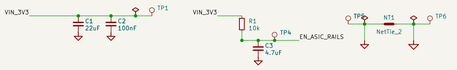
FRONTEND MATING (TOP FACE)



ASIC PORTS – TWO ASIC BOARD PAIRS (J3 TOP, J4 BOTTOM, SAME SPOT)



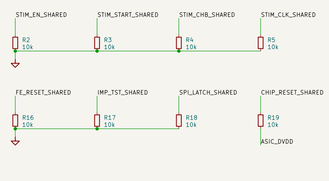
VIN_3V3 ENTRY, DELAYED ENABLE, RETURN BOND



1. Function: C1/C2 hold VIN_3V3 for the converter and regulators; it arrives on J2.68–70, the three Frontend supply contacts. R1/C3 delay EN_ASIC_RAILS by about 20 ms after VIN_3V3, so all nine ASIC rails start after the frontend 1.5 V bank supply. NT1 is the only AGND–GND bond for all four ASIC boards (the ASIC boards must not add their own).

2. Source / changes: Claude design, no manufacturer reference circuit. Delay: tau = R1 x C3 = 47 ms; LT3094 EN/UV turns on at 1.26 V, so t = 47 ms x ln(3.3/2.04) = 23 ms; LT3042 threshold assumed similar (to confirm). Nine EN/UV pin currents x 10k must stay well below 1.26 V. The frontend VCCO_1V5 start time is not measured. NT1 follows the LDO V13 NT100 single-bond idea: the ASIC boards carry AGND (42/44) and DGND (46/48) separately on J3/J4 and no part on them joins the two (read from the V6 board files) (checked in its PCB file). V34: meeting with the ASIC designer, 12 Nov 2026 (2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), answer B3, confirms two grounds joined at one point on this board, far from the chip; NT1 stays the only bond. C901/C1001 return to AGND and the LT3582 capacitors to GND, so part of the converter ripple still crosses NT1 (review items LDOA–10, LDOA–12; not changed in the schematic: layout V13 found no legal site for plane vias at NT1). Unverified: ripple between AGND and GND at the ports, and whether VIN_3V3 sags at the J2 contacts (0.9 A for three).

ASIC CONTROL SAFE IDLE



1. Function: Hold every FPGA-to-ASIC control line at its safe level while the FPGA is unconfigured: STIM_EN/START/CHB/CLK, FE_RESET, IMP_TEST and SPI_LATCH low; CHIP_RESET high (to ASIC_DVDD), so the ASICs stay in reset until the FPGA releases them. About 150 uA per line when driven to the other level.

2. Source / changes: Claude design from Gerald's firmware (Archive/.../07_FPGA_512_Reference/repository/fpga): top_custom.v section 3b drives CHIP_RESET high for 2^20 cycles (~65 ms) at boot then low (active high); line 264 holds FE_RESET low ("amplifier reset released; normal operation"); STIM_* held low; custom_board.qsf line 100 leaves IMP_TEST an input with PULLMODE=DOWN; spi_programmer.v keeps SCLK idle low and pulses LATCH high. Needed because the Frontend ties PUDC_B high (R25 470R), disabling FPGA pull-ups during configuration (AMD UG470, PUDC_B). V23 adds R16–R23 to V22's R2–R5. V28 (Howard, 9 Nov 2026) had dropped the four clock pulls R20–R23; V34 brings them back with four more for the top chips, drawn in the block ASIC CLOCK AND TOP-CHIP SPI SAFE IDLE, READ/SYNC are ASIC outputs and get no pull. Unverified: ASIC input leakage and thresholds: reset polarity is from firmware, not an ASIC data sheet.

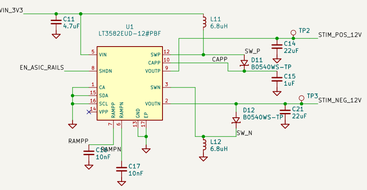
AC_IN DEFAULT



1. Function: R6 holds the analog AC_IN input of all eight ASICs at AGND; TP7 lets a bench source inject a test signal (0–1.5 V nominal) through the shared node.

2. Source / changes: Claude design, The Frontend has no AC_IN source and no connector contact for it (V43), so AC_IN is handled here. Layout 13: in version B these parts sit on the router, so AC_IN needs no J5 contact. Nominal 0–1.5 V from ASIC_REFERENCE.md section 4 (relayed designer answer). Gerald, 9 Nov 2026: "0V is the right state. No need to care about input impedance or anything else."

STIMULATION SUPPLY – VIN_3V3 TO +/–12 V (LT3582-12) THEN +/–8.06 V LDOs



1. Function: LT3582–12 makes STIM_POS_12V (+12 V, boost through D11 to CAPP and its output-disconnect switch to VOUTP) and STIM_NEG_12V (–12 V, single-inductor inverter through D12) from VIN_3V3, as pre-regulators for the stimulation LDOs. Fixed +/–12 V with internal feedback and compensation; starts with EN_ASIC_RAILS (SHDN on above 1.1 V). About 10 mA per output.

2. Source / changes: ADI LT3582/LT3582–5/LT3582–12 data sheet 3582512fb, Figure 13 p.22 (+/–12 V from a single 5 V input). Pin Functions p.8, Choosing Inductors p.18, Capacitor and Diode Selection p.19. https://www.analog.com/media/en/technical-documentation/data-sheets/3582512fb.pdf . Changed from Figure 13: input 3.3 V instead of 4.5–5.5 V (allowed 2.55–5.5 V: max load at 3.3 V about 40 mA per output from the p.19 equations, need about 10 mA); C1 –> C11 4.7 uF at 10 V 0603; C2 –> C21 22 uF 25 V 0805 and C3 –> C14 22 uF 25 V 0805 (about 6–7 uF effective at 12 V, inside the 2.2–10 uF range); C4 –> C15 1 uF 25 V on CAPP: C5/C6 –> C16/C17 10 nF; L1/L2 6.8 uH (Coilcraft XPL2010-682 in the figure) –> L11/L12 TDK VLS201612HBX-6R8M–1, LCSC C5015945 (V35: before: Murata DFE201610E-6R8M=P2, which JLC does not list): 6.8 uH +/–20 %, Isat 0.91 A, Itemp 0.87 A, 0.468 ohm max, 1.2 mm high (TDK VLS201612HBX–1 data sheet 20180709, p.1 table and p.3 drawings, https://datasheet.lcsc.com/datasheet/pdf/dce85b551190e1e89e74a90d08568b6cf.pdf). Needed (p.3 current limits, p.18 equation IPK = ILIMIT + VIN x TOS / L): 0.48 A peak in L11 and 0.78 A in L12, 0.80 A with L 20 % low and 3.6 V in, TDK land pattern 0.5 x 1.6 mm pads, 2.0 mm over all; board pads 0.55 x 1.6 mm, 2.0 mm over all; D1/D2 Diodes Inc. B0540WS–7 in the figure –> D11/D12 MCC B0540WS–TP, JLC C668860 (V35: the Diodes part, JLC C98664, cannot be ordered at JLC): same type and case, 40 V, 0.5 A average, 5 A surge, 0.55 V max at 0.5 A, SOD–323 (MCC B0540WS data sheet Rev.3–2 07102019, p.1, https://datasheet.lcsc.com/datasheet/pdf/231ff9d51761a20ad69da348e7944c17.pdf). Needed; 12.1 V reverse on D11 and 15.4 V on D12, average = the load of about 10 mA, peak as the inductors. Different from the Diodes part: leakage limit 80 uA at 40 V against 5 uA at 30 V (0.8 % of a 10 mA load); neither sheet gives a repetitive-peak rating, MCC leads 2.30–2.70 mm over all and 0.25–0.40 mm wide; board pads 0.60 x 0.45 mm, 2.70 mm over all; SHDN to EN_ASIC_RAILS instead of an input logic signal; CA/SDA/SCL to GND, VPP open (I2C/OTP unused on the –12 version). V24 replaces the TPS65130 (Howard, 8 Nov 2026: smaller parts) and removes R11–R15, C12, C13, C18–C20, C22. V34: the +/–12 V setting is unchanged; the LDOs after it are now set to +/–8.06 V. V35 (13 Nov 2026): part fields only, no net and no copper changed. U1 LT3582EUD–12#PBF is JLC C670975: 0 in JLC stock, so it comes through JLC Global Sourcing; only the –12 version works here. JLC stock read 13 Nov 2026, 00:12 CST: C668860 5,169 orderable, C5015945 1,170 orderable (10 of each needed for 5 boards), C670975 0, Unverified: ripple after layout, start–up current at the J2 supply contacts, diode leakage when warm; nothing is measured yet.

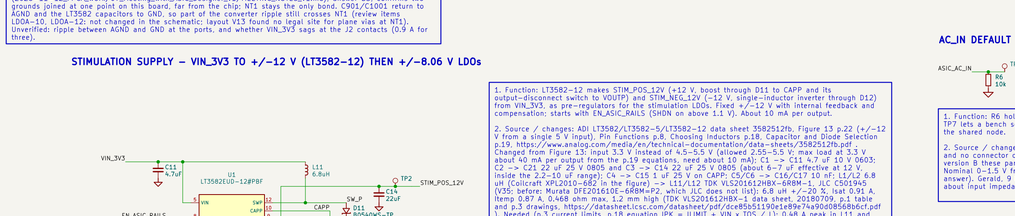
STIM SUPPLY SELECT



1. Function: JP1/JP2 connect the on-board LT3582–12 outputs to the stimulation LDO inputs. Cut both to feed STIM_PRE_POS/NEG from a bench supply on TP8/TP9 (return: TP6 AGND), as LDO Board A did.

2. Source / changes: Claude design. LDO Board A took INPUT_VDD10/INPUT_VSS10 from outside (Reference/LDO Board A 715/J8). Limits: LT3042 IN <= 20 V, LT3094 IN >= –20 V (ADI data sheets, Absolute Maximum Ratings); keep >= 1.5 V headroom over the +/–8.06 V outputs (V34), Leave U1 unfitted, or its outputs open, when external +/–V is used. Unverified: external source noise.

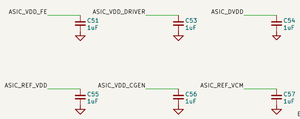
ASIC RAILS – THREE LT3042 (ANALOG / REFERENCE) + FOUR TPS7A20 CELLS FROM VIN_3V3



1. Function: Seven regulators make the ASIC rails from VIN_3V3; each rail is shared by all eight ASICs through the two ASIC ports. LT3042 (U2 VDD_FE, U6 REF_VDD, U8 REF_VCM 0.75 V): Vout = 100 uA x RSET, TPS7A2015 (U3 VDD_ADC, U4 VDD_DRIVER, U5 DVDD, U7 VDD_CGEN): fixed 1.50 V. All start from EN_ASIC_RAILS (TPS7A20 EN high above 0.9 V max, LT3042 EN/UV above about 1.24 V).

2. Source / changes: LT3042 cells Inherited from Gerald LDO Board A (Reference/LDO Board A/PCB.kicad_sch): IN, EN/UV and PGFB at the input, ILIM to ground, OUT=OUTS, CIN/COUT 10 uF 10 V 0603, CSET 1 uF, PG open, RSET 15k/7.5k: ADI LT3042 data sheet, Typical Application p.1 and Pin Functions, https://www.analog.com/media/en/technical-documentation/data-sheets/lt3042.pdf. Claude changes: EN/UV to EN_ASIC_RAILS; one regulator per rail for eight ASICs, V24 (Howard, 8 Nov 2026, "find smaller parts"): U3/U4/U5/U7 changed from LT3042 3x3 DFN cells to TI TPS7A2015PDQNR 1x1 mm X2SON (TI SBVS338H Rev H, Jul 2024: Pin Functions p.4 DQN 1 OUT/2 GND/3 EN/4 IN/5 pad: Recommended Operating Conditions p.5 CIN 1 uF, COUT 1–200 uF with >= 0.47 uF effective; Electrical Characteristics p.7 VEN(HI) 0.9 V max: https://www.ti.com/lit/ds/symlink/tps7a20.pdf): their RSET/CSET (R301–R701, C303–C703) are removed and CIN/COUT are 1 uF 10 V 0402. Trade-off: 7 uVrms against 0.8 uVrms (LT3042), accepted by Gerald 9 Nov 2026 ("I think its okay to do that"). V34 (12 Nov 2026): every regulator now returns to the ground its rail is decoupled to on the ASIC boards: U3 (VDD_ADC) with C301/C302 moved from GND to AGND, and U5 (DVDD) with C501/C502 moved from AGND to GND: the other five were already on AGND. Basis: ASIC boards V6 decouple DVDD to DGND (C7, C14, C61, C62) and every other rail to AGND (read from both V6 board files, 12 Nov 2026); meeting with the ASIC designer, 12 Nov 2026 (2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), answer B3 (two grounds, joined at one point on this board): review items LDOA–11 and LDOA–16. REF_VCM stays LT3042 because 0.75 V is below the TPS7A20 0.8 V minimum. Loads of REF_VDD/REF_VCM unknown.

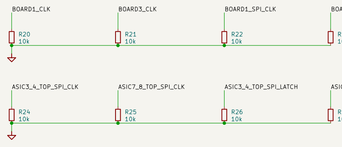
RAIL CAPACITORS AT THE ASIC PORTS



1: Function: C51 and C53–C57 put 1 uF on six ASIC rails where they leave for the ASIC stacks (C51, C53, C55, C57 at the J3/J4 ports; C54 beside U5; C56 beside U7). Each returns to the ground the ASIC boards decouple that rail to: C54 (DVDD) to GND, which is the ASIC boards' DGND; the other five to AGND.

2. Source / changes: Claude design (7 Nov review, V20), no manufacturer reference circuit. Changed in V34: C54 moved from AGND to GND, because ASIC boards V6 decouple DVDD to DGND (C7, C14, C61, C62) and every other rail to AGND (read from both V6 board files, 12 Nov 2026): review items LDOA–11 and LDOA–16, V28 (Howard, 9 Nov 2026): C52 (VDD_ADC), C58 (VDD10) and C59 (VSS10) removed; those rails keep their regulators' output capacitors. All are 1 uF 10 V 0402, Samsung CL05A105KP5NNNC, JLC C14445 (JLC stock read 12 Nov 2026, 19:37 CST). Unverified: the capacitor values on the ASIC boards (not yet given by the designer), and the regulators' stability with this extra capacitance at the end of the trace (LT3042/LT3094 are stable with larger output capacitance per their data sheets).

ASIC CLOCK AND TOP–CHIP SPI SAFE IDLE (V34)



1. Function: R20–R27 hold the FPGA-to-ASIC clock and latch inputs that had no pull at a low level while the FPGA is unconfigured or a line is undriven: the fast clock CLKH of both ports (BOARD1_CLK, BOARD3_CLK), the bottom chips' SPI clocks (BOARD1_SPI_CLK, BOARD3_SPI_CLK) and the top chips' own SPI clocks and latches (ASIC3_4_TOP_SPI_CLK/LATCH, ASIC7_8_TOP_SPI_CLK/LATCH): About 150 uA per line while it is driven high.

2. Source / changes: Claude design, no manufacturer reference circuit: same part as R2–R5/R16–R19: 10k 1 % 0402, Uniroyal 0402WGF1002TCE, JLC Basic C25744 (16.2 million orderable, JLC stock read 12 Nov 2026, 19:37 CST). Reason: the ASIC designer said the chip pads have no internal pull resistors (meeting with the ASIC designer, 12 Nov 2026 (2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), answer C2), so this board must set every level; review items LDOA–49 and SYS–28. R20–R23 return the four clock pulls that V28 removed (Howard, 9 Nov 2026): R24–R27 are new for the top chips' own SPI clock and latch of pin map V6 (port pins 72 and 70). Idle–low follows Gerald's firmware (spi_programmer.v keeps SCLK low and pulses LATCH high). Not pulled: the 16 SPI data inputs (STIM_CHIPx_SPI_DL/DR, ASICn_SPI_DL/DR): they are only taken in on an SPI clock edge and a latch pulse, and both are now held low. Unverified: ASIC input leakage and thresholds: the designer gave no preferred idle level (answers C1, C3, C4: "should not matter much"). All eight are placed and routed in layout V13. R23 sits on the top side beside R4, tapped at contact J2.13; the other seven are on the bottom side.

5 V PROVISION (UNUSED)



1. Function: Brings the Frontend's 5 V input (VIN_5V) onto this board on one contact, J1.32, and ends it on the test pad TP10. Nothing on this board uses it: it is a provision, so the regulation can be changed later without a new connector map.

2. Source / changes: Claude design, no manufacturer reference circuit, Howard, 12 Nov 2026 (decision 4 after the meeting with the ASIC designer, 12 Nov 2026 (2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), where the designer suggested also sending 5 V to the LDO board). Contact from pin map V6 (12 Nov 2026; 2026-11-12_FPGA_Work/09_Pin_Map_V6/PIN_MAP_V6.csv and ASIC_PORT_PINS_V6.csv, working copy /Users/howardwang/kicad_work/pinmap_v6/deliver/pin_map_v6.json; published as the working map, not frozen): J1.32 = VIN_5V with J1.30 (new GND) and J1.34 (GND) as row neighbours, so a solder bridge cannot put 5 V on a 1.5 V line. One DF40 contact is rated 0.3 A (Hirose DF40 catalog, https://datasheet.lcsc.com/datasheet/pdf/145ab5d527b34eafa7a47375cb561c6a.pdf?productCode=C3640153). No capacitor: the net has no load here. Layout V13: 0.20 mm on L1 between the two J1 contact rows, 0.29 mm from the Inner ends of the 1.5 V contacts 34–68. Unverified: the Frontend does not drive J1.32 yet (its next revision does): the load of any later 5 V use.

1. Function: Mate the Artix Frontend bottom-side receptacles J1/J2 (DF40C–70DS) with two DF40C–70DP plugs on this board's top face, 1.5 mm stack, pin n to pin n. J1 runs along the left long edge and J2 along the right long edge, parallel, pin 1 at the ASIC end. Carries the 120 ASIC signals (J1 60, J2 60), VIN_3V3 (J2.68–70, three 0.3 A contacts), VIN_5V (J1.32, one contact, an unused provision), 15 GND contacts and one spare (J1.36, open on this board).

2. Source / changes: Claude design, pin map V6 (12 Nov 2026; 2026-11-12_FPGA_Work/09_Pin_Map_V6/PIN_MAP_V6.csv and ASIC_PORT_PINS_V6.csv, working copy /Users/howardwang/kicad_work/pinmap_v6/deliver/pin_map_v6.json; published as the working map, not frozen). Built on pin map V5 (layout V11 lane plan: Howard, 10 Nov 2026: one Frontend connector on each long edge, every ASIC group whole on one connector). Changed from schematic V30 (map V5): 12 contacts that were spare now carry top–chip data and READ lines (J1.29, J1.31, J1.33, J1.35, J1.37, J1.39, J1.43, J1.45, J1.47, J1.49, J2.40, J2.42); 22 contacts keep their wire but carry another signal (top–chip SPI data, the top chips' own SPI clock and latch, top–chip inputs moved along the bus): J1.32 = VIN_5V; J1.30 = GND, so both row neighbours of the 5 V contact (J1.30, J1.34) are ground; J1.36 stays open here. Reason: meeting with the ASIC designer, 12 Nov 2026 (2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), answers A1 and A2 (pin order final: top chips keep their own SPI clock and latch), and Howard's decisions of 12 Nov 2026. Unchanged: VIN_3V3 on J2.68–70. GND on 14/15/27/34/41/50/66 of both. Hirose DF40 catalog, June 2024 (LCSC copy), URL https://datasheet.lcsc.com/datasheet/pdf/145ab5d527b34eafa7a47375cb561c6a.pdf?productCode=C3640153 . DF40C–70DP C2692097 / 70DS C294854 (JLC, read 2 Nov 2026). Unverified: the Frontend has not adopted map V6 yet (its next revision does): two rigid DF40 pairs between two boards (the DF40 drawing, sheet 3 item 4, advises one pair per board pair).

1. Function: Two ASIC board pairs sandwich this board. J3 (plug, top face) feeds the pair above: the bottom ASIC board (ASIC5/ASIC6) and, through its T_ pass–through lines, the top ASIC board (ASIC7/ASIC8). J4 (receptacle, bottom face, same XY) feeds the pair below the same way (ASIC1/ASIC2, then ASIC3/ASIC4). Each port also carries the nine ASIC rails, AC_IN, 2 AGND and 2 DGND contacts.

2. Source / changes: ASIC boards V6, bottom and top (12 Nov 2026; built on the designer's updated pair): J3 and J4 carry the same name on each pin number of both boards (pin map V6 check: 0 differences on 80 pins x 4 connectors). Pin assignment: pin map V6 (12 Nov 2026; 2026-11-12_FPGA_Work/09_Pin_Map_V6/PIN_MAP_V6.csv and ASIC_PORT_PINS_V6.csv, working copy /Users/howardwang/kicad_work/pinmap_v6/deliver/pin_map_v6.json; published as the working map, not frozen). Changed from schematic V30: the designer's final 80–pin order is the old order with the west T_ bus moved by four pins, so pins 2/4/6/8 now carry T_READ[2]. T_D<8>[2].
T_D<7>[2], T_D<6>[2] and pins 10–22 the lines that sat four pins earlier; pins 24/26/28/30/32/58 carry top–chip lines that had no contact: pins 16/18 and 66/68 are the top chips' own SPI data, pin 70 their SPI latch and pin 72 their SPI clock (meeting with the ASIC designer, 12 Nov 2026
(2026-11-12_FPGA_Work/07_Meeting_With_ASIC_Designer_V1/MEETING_NOTES_V1.md), answer A2: not shared with the bottom chips). Pins 3 and 5 (DISC_OUT) stay unconnected: not needed (answer A3). CLKH stays one line per port (BOARD1_CLK, BOARD3_CLK). Unverified: the designer's own check of his board pair: data bit order and READ/SYNC timing (firmware only).

Pin map V6: Frontend J1/J2 on top; ASIC ports J3 top, J4 bottom.
3x LT3042 + 4x TPS7A20 rails: LT3582–12; LT3042/LT3094 +/–8.06 V: AGND–GND tie only at NT1.
Layout V14 (copper of layout V13; part fields only). Input: VIN_3V3 from Frontend J2.68–70. VIN_5V on J1.32.

Rice – FPGA PCB workspace

Sheet: /
File: FPGA_LDO_A_layout_V16.kicad_sch

Title: FPGA–LDO schematic V35 – Option A, one router/LDO board

Size: User	Date: 2026-11-13	Rev: V35
KiCad E.D.A. 10.0.6		Id: 1/1